Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

WEEKLY LESSON PLAN

Department: CSE
Year / Semester: 2025-26 / Winter
Course Code / Course Name: 10211CS225 / Problem Solving and Testing using Java

UNIT II – Algorithmic Thinking & Competitive Problem Patterns **Course Outcome:** Apply algorithmic patterns to develop optimized solutions.				
Week	**Session**	**Topics to be Covered**	**Problems to Practice (Hands-On)**	**Platform**
Week 6	Session 1	KMP Algorithm	Task1:https://www.hackerrank.com/challenges/string-similarity/ Task 2:https://leetcode.com/problems/repeated-substring-pattern/	Leetcode,Hackerrank
	Session 2	KMP Applications	Task3:https://www.hackerrank.com/challenges/two-strings/ Task4:https://leetcode.com/problems/rotate-string/	
	Session 3	Boyer Moore Algorithm	Task 5:https://www.hackerrank.com/challenges/mars-exploration/ Task6:https://leetcode.com/problems/find-all-anagrams-in-a-string/	
	Session 4	Palindrome Techniques	Task7:https://www.hackerrank.com/challenges/palindrome-index/problem Task8:https://leetcode.com/problems/find-the-index-of-the-first-occurrence-in-a-string	
	Session 5	Manacher Algorithm for finding longest palindromic substring	Task 9:https://leetcode.com/problems/longest-palindromic-substring/ Task10:https://www.hackerrank.com/challenges/circular-palindromes/	